Ans4

Deadlock is a condition in which processes are blocked forever because each is waiting for a resource held by another process. Coffman conditions explain when deadlock can occur:

1. Mutual exclusion: resource cannot be shared.
2. Hold and wait: process keeps one resource while asking another.
3. No preemption: resource cannot be forcefully taken.
4. Circular wait: chain of processes waiting for each other.

If all four happen together, deadlock is possible. Example: P1 holds printer and wants scanner; P2 holds scanner and wants printer. Neither proceeds. OS can handle deadlocks through prevention, avoidance, or detection/recovery methods.

Ans5

Virtual memory provides each process a large logical memory space, even when physical RAM is limited. The OS stores less active pages on disk and loads required pages into RAM on demand. Paging divides logical memory into pages and physical memory into equal-sized frames, then maps them via page table.

Main benefits are improved multiprogramming, process isolation, and reduced external fragmentation because contiguous physical allocation is not required. But if page faults happen too often, performance drops due to disk latency. So page replacement policy and frame allocation are important for efficiency.

Ans6

Synchronization is required when multiple threads/processes access shared data concurrently. Without synchronization, race conditions occur and outputs become non-deterministic. Critical section is the part of code where shared data is accessed and must be protected.

Mutex is a lock for mutual exclusion: one thread enters, others wait. Semaphore is a signaling counter and can allow one or more threads depending on binary or counting type. Semaphores are useful in producer-consumer control and resource pools. Proper synchronization ensures correctness, consistency, and safe concurrent execution.

Ans7

Fragmentation means wasted memory due to allocation patterns. Internal fragmentation happens when allocated block is larger than requested memory, leaving unused bytes inside that block. External fragmentation occurs when free memory is scattered into small holes and large contiguous allocation fails.

Internal fragmentation can be reduced by using better allocation sizes, while external fragmentation can be reduced using compaction or paging. Paging is especially effective against external fragmentation, though some internal waste may still exist in the last page.